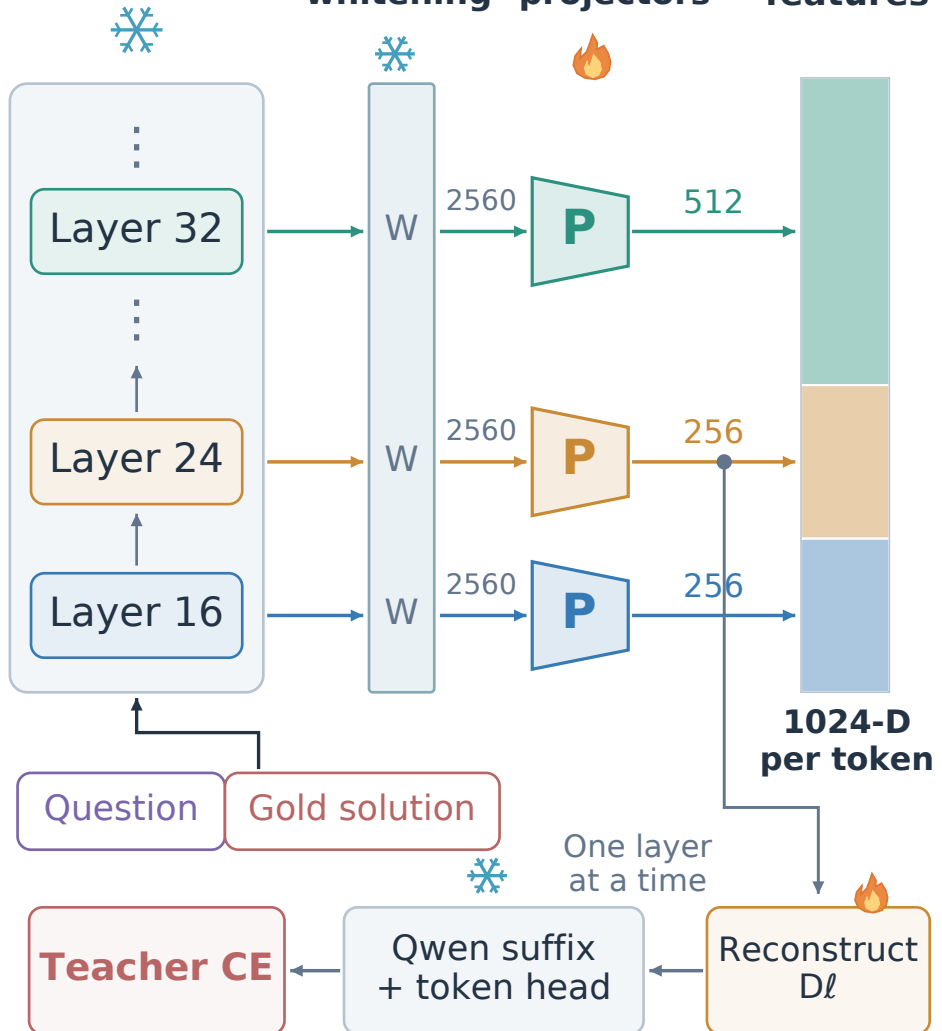


1 Representation learning

Qwen3-4B **Covariance whitening** **Learned projectors** **Concatenate features**



Freeze this representation for flow training.

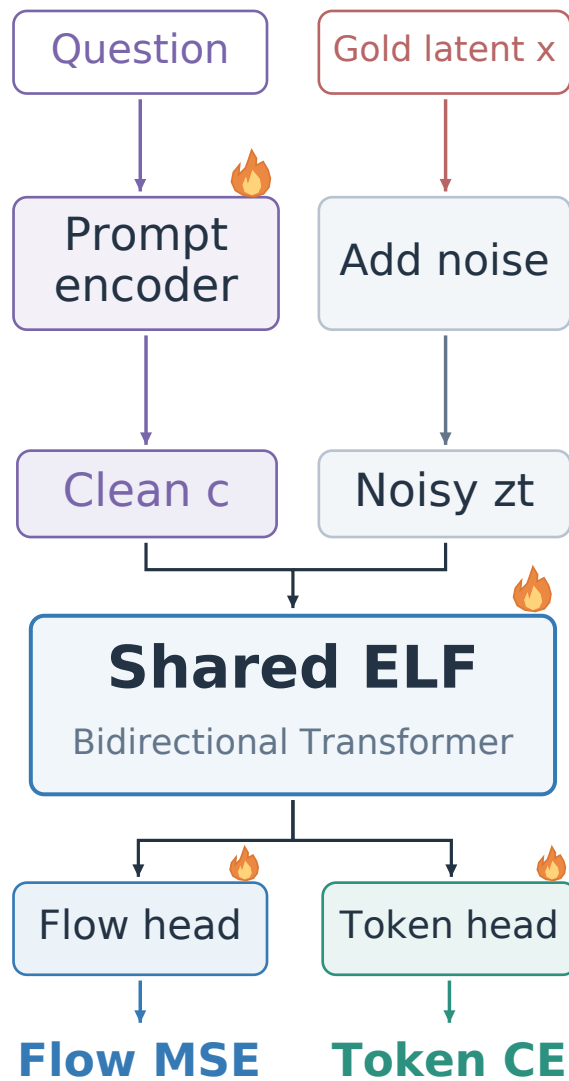


Frozen



Trained

2 Flow training

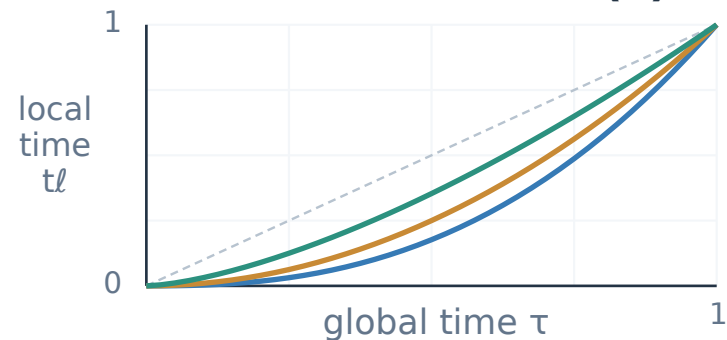


Teacher prompts $\rightarrow$ MSE
warm start $\rightarrow$ joint training

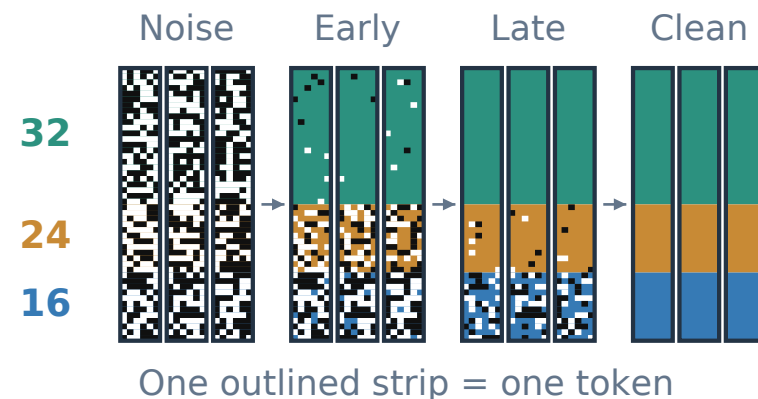
3 Async inference

Encode question once $\rightarrow$ cache c

Local clocks: $t_l = f_l(\tau)$



— $\gamma_{16}=2.5$ — $\gamma_{24}=2$ — $\gamma_{32}=1.5$



Joint ELF calls + Euler updates

ELF + token head $\rightarrow$ answer

Noise contrast exaggerated for illustration